



Finetuning Architecture

SOUTHWEST RESEARCH INSTITUTE®

Andrés Muñoz-Jaramillo

andres.munoz@swri.org

Commissioned and funded by NASA's Office of the Chief Science Data Officer (OCSDO)

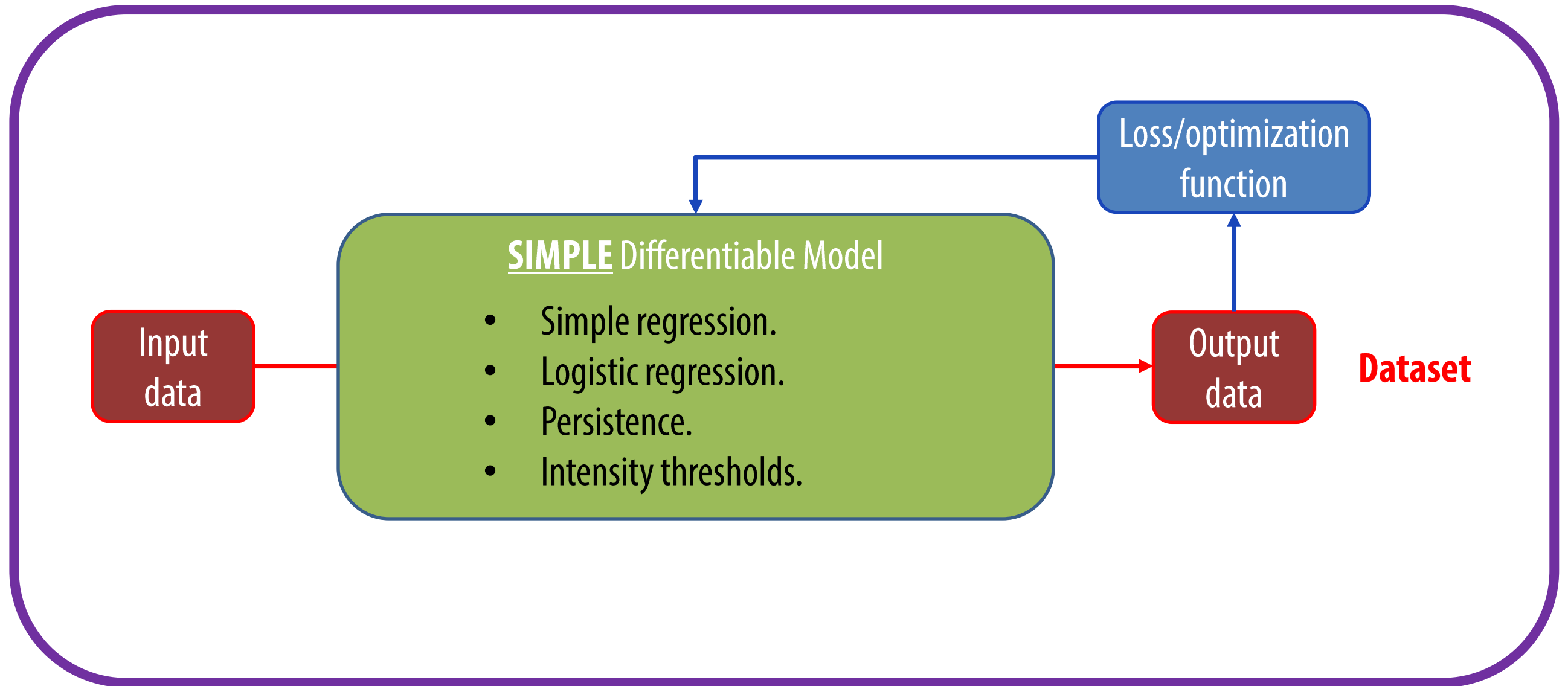


SOLAR SYSTEM SCIENCE & EXPLORATION

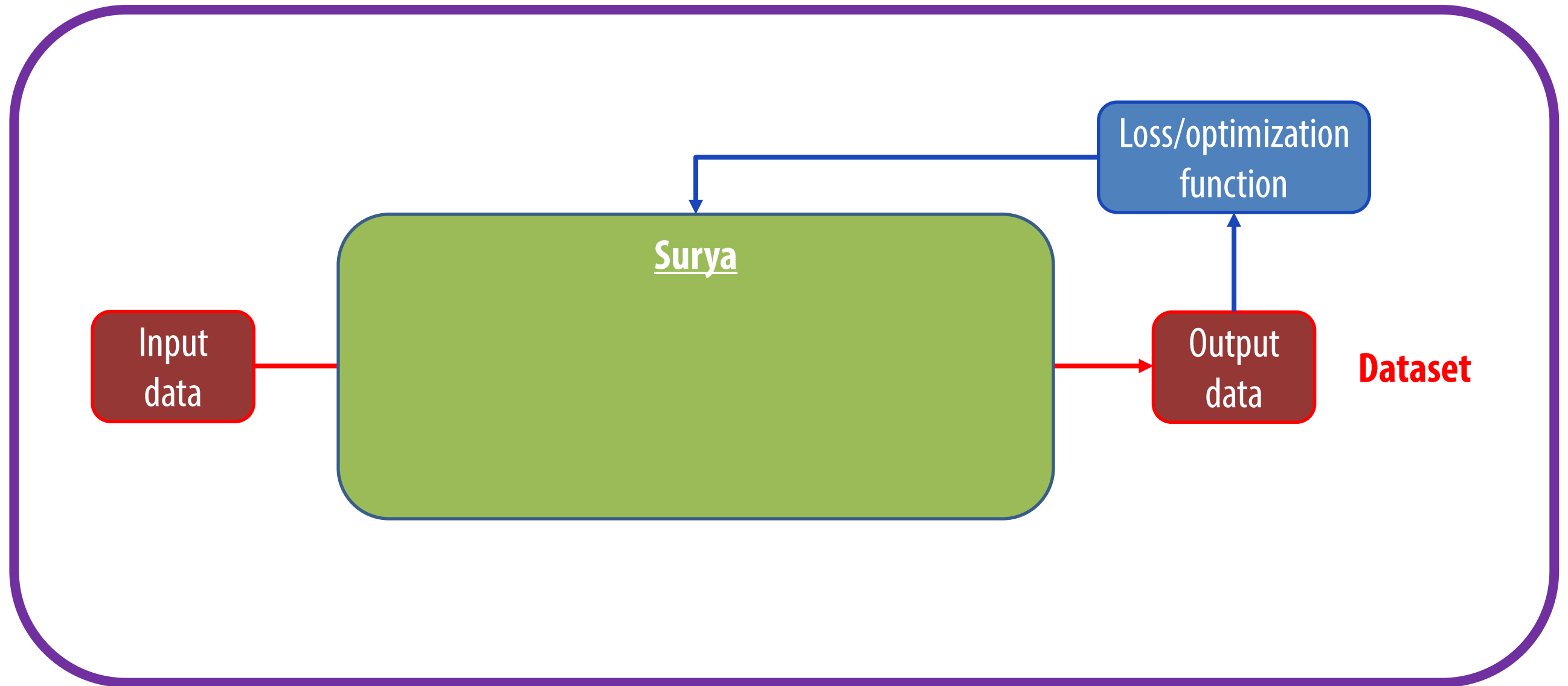
Some things to keep in mind

- Let's keep an eye on space and the cache of Surya stacks. We are exploring what it means to live in the future!
- Please keep me informed about need for more compute. Our potential capability is modest, but we can bring more online.
- If you are a virtual participant, use your breakout rooms and voice.
- The finetuning template is online, but be careful updating your branch.

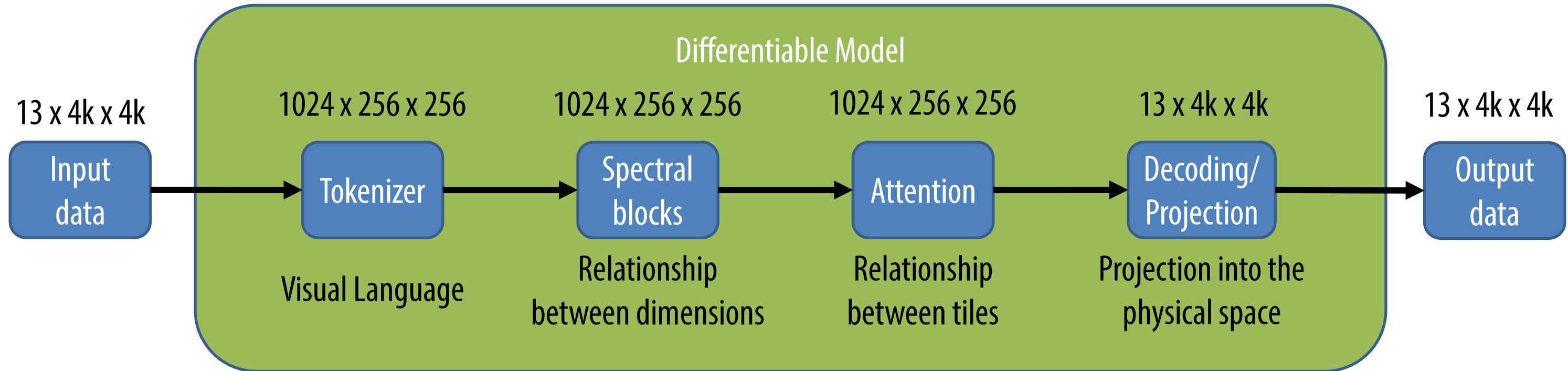
The training loop: PyTorch lightning



The training loop: PyTorch lightning

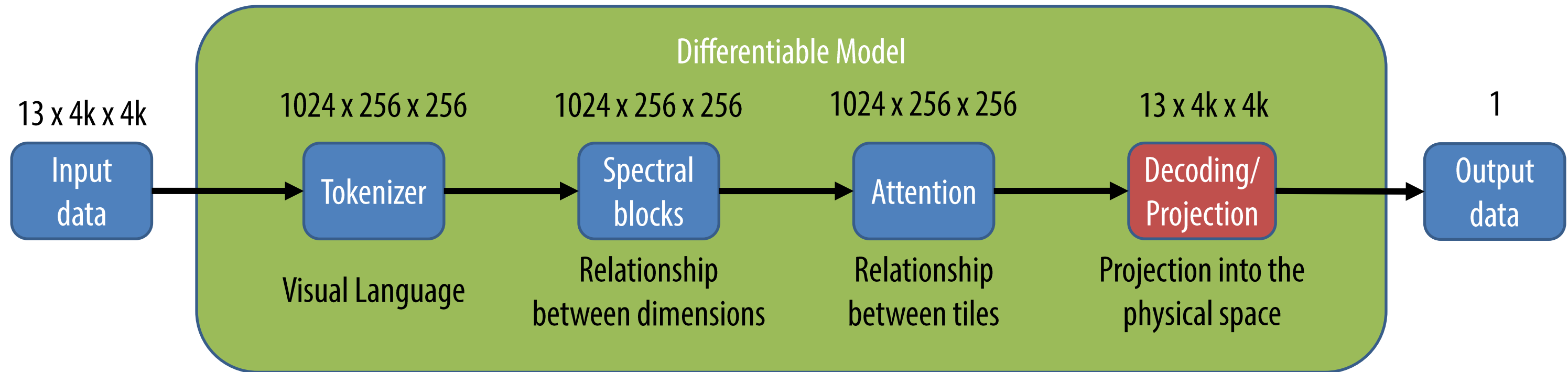


Transformation and projection



- Start by making dimensions match!

1D Application

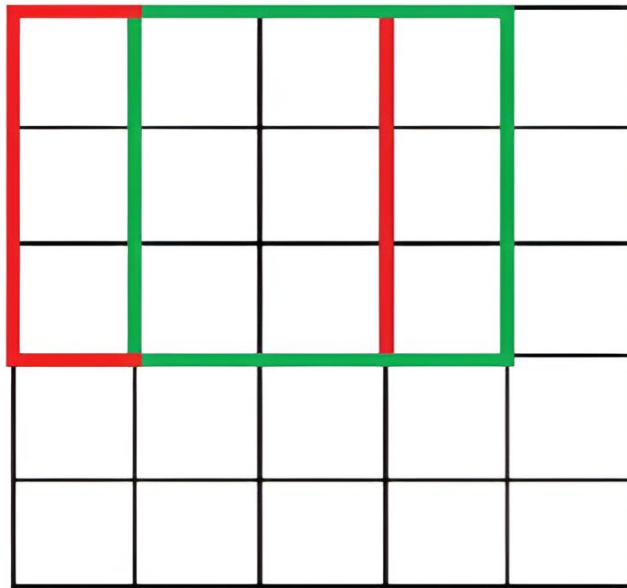


- Start by making dimensions match!

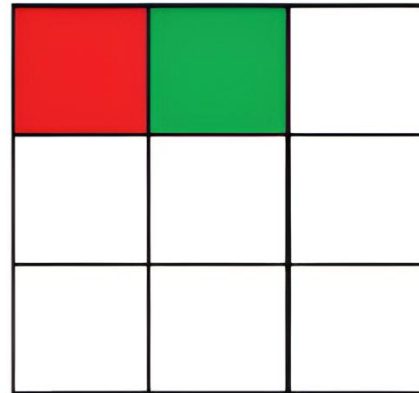


Downscale

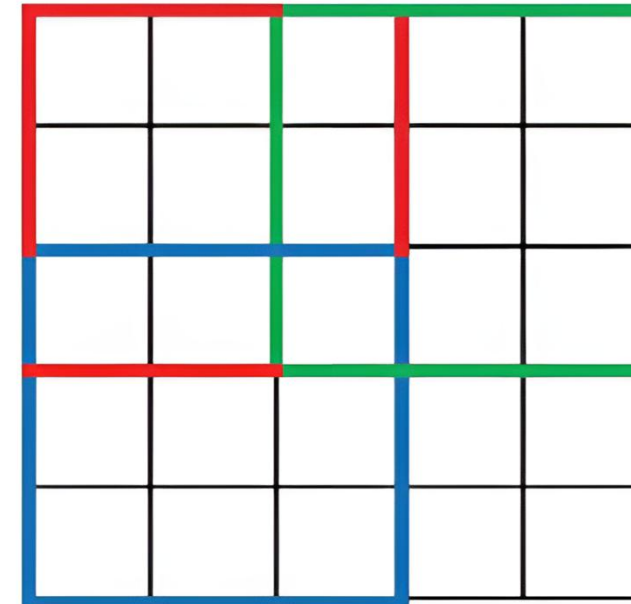
Convolution
with Stride=1



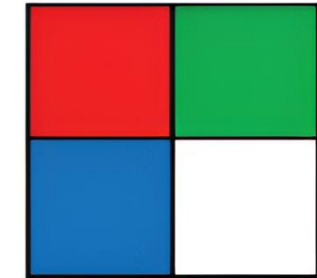
Output



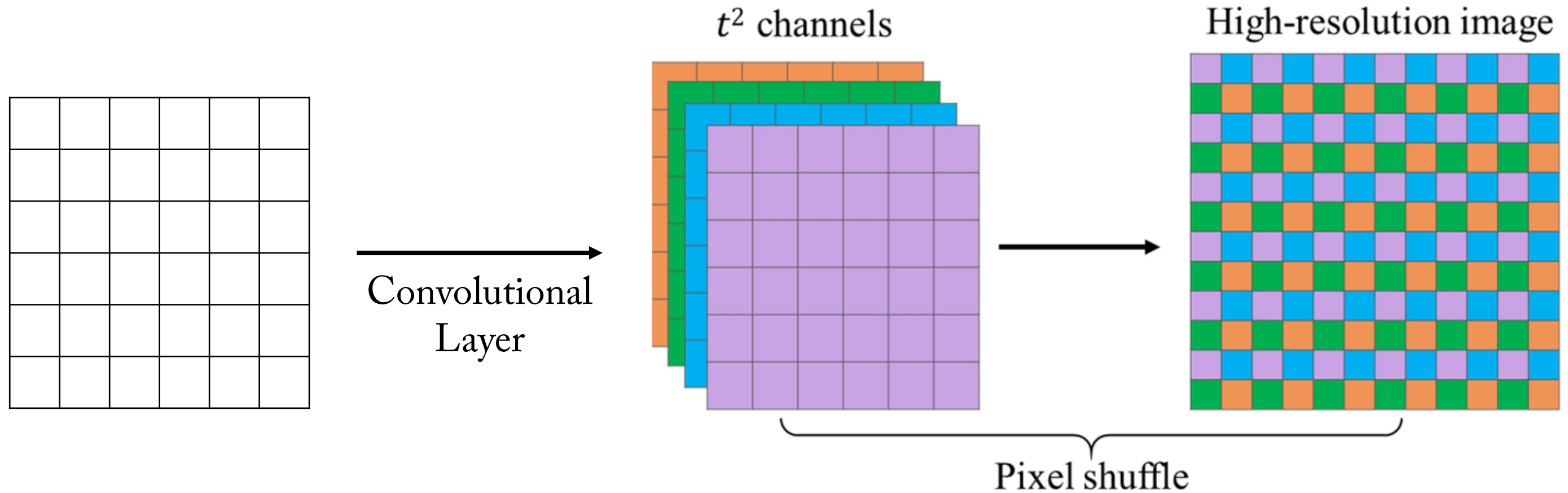
Convolution
with Stride=2



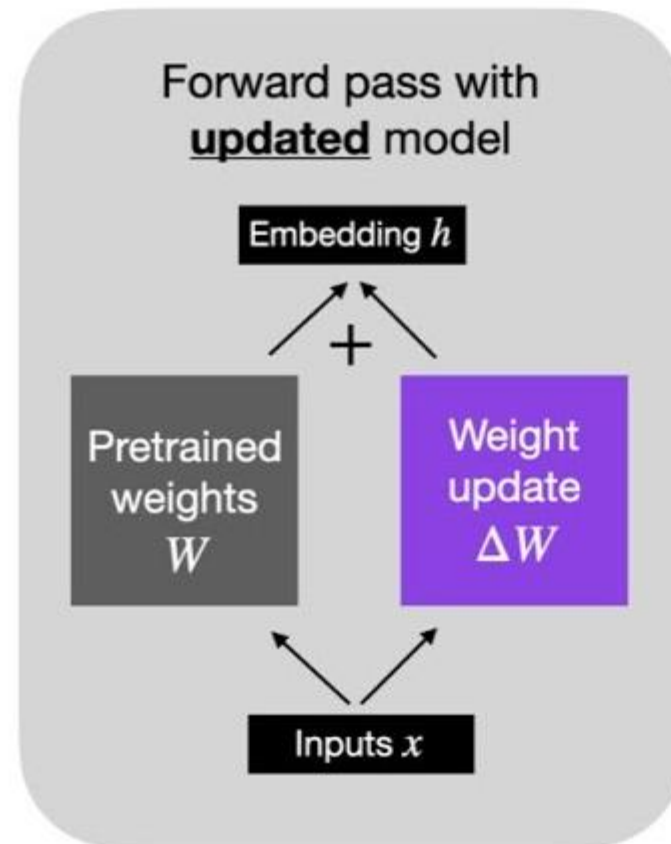
Output



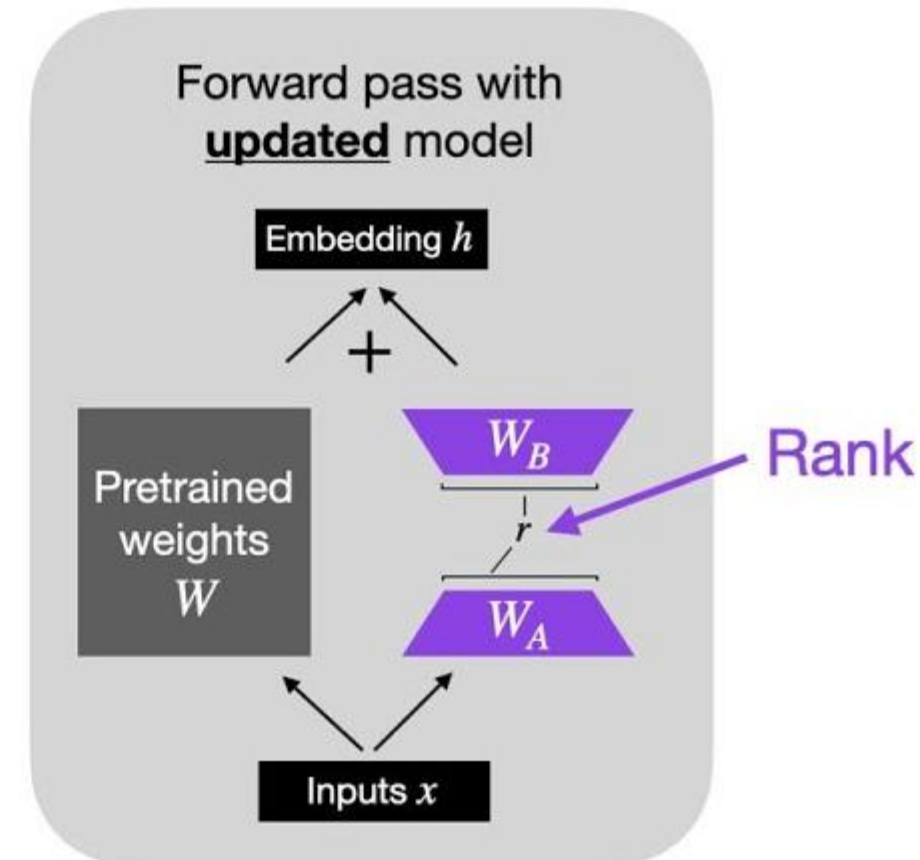
Upscale



Low Rank Approximation - LoRA



Regular finetuning



Low-rank approximation

Surya may work out of the box on different resolutions

Thanks Zhenduo!